# XUNHUI WU

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## SUMMARY

Ph.D. candidate in Neuroscience at Northwestern University with expertise in biomedical imaging, *in vivo* pharmacology, and signal processing. Specialized in dissecting dopamine and acetylcholine signaling in basal ganglia circuits relevant to neuropsychiatric and movement disorders. Proficient in Python and MATLAB for building pipelines to analyze large-scale *in vivo* imaging datasets. Experienced in validating medical technologies (drug delivery systems) and modeling disease states. Recipient of competitive NIMH funding, seeking to apply rigorous R&D skills.

## TECHNICAL SKILLS

**Biomedical Imaging*:*** 2-photon Calcium Imaging, Fiber Photometry, Neural Signal Processing, Time-Series Analysis.
**Pharmacology and Behavior:** *in vivo* Pharmacology, Behavioral Assays (Operant conditioning, Open Field, CPP), Stereotaxic Surgery.
**Computational**: Python (NumPy, Pandas, Scikit-learn), MATLAB, SQL.
**Data Science**: High-dimensional Data Analysis, Pipeline Development, Machine Learning, Data Visualization, Reproducible Research Workflows.
**Biology*:*** Parkinson's and Alzheimer's Disease Models, RNA Extraction, Flow Cytometry, Biobanking.
**Tools*:*** Git, Adobe Illustrator, ImageJ.

## EDUCATION

**Northwestern University**; Feinberg School of Medicine; Chicago, IL — September 2021 – Present; Expected Graduation: June 2027
Ph.D. in **Neuroscience**
**Northwestern University** — September 2020 – August 2021
Master of Science in **Neurobiology**
**University of Toronto** — September 2015 – June 2020
Bachelor of Science in **Molecular Genetics**, **Physiology** with High Distinction

## PROFESSIONAL EXPERIENCE

**Graduate Student Researcher, Laboratory of Jones G. Parker, PhD •** Neuroscience, Northwestern University; Chicago, IL
Mar 2022 – present
Thesis: Role of striatal μ-ORs in striatal coding of compulsive opioid use.

- Characterized neural substrates associated with opioid abuse through *in vivo* calcium imaging.
- Dissected the necessity of μ-opioid receptors in reinforcement behaviors using site-specific genetic manipulations and pharmacological challenges.
- Engineered automated signal processing pipelines (Python/MATLAB) to process terabytes of high-resolution imaging data. Implemented signal demixing, event detection, and behavioral integration to decode cellular networks under pharmacological influences.
- Used multiplexed fiber photometry to record dopamine and acetylcholine signaling at different receptors following treatment of antipsychotic and related drugs.

**Teaching Assistant •** Northwestern University, Interdepartmental Neuroscience Program
Sep 2022 – Jun 2023

- TA for neuroscience and biology courses; led discussions and labs.

**Master's Student, Laboratory of Marco Gallio, PhD •** Neurobiology, Northwestern University; Evanston, IL
Sep 2020 – August 2021

- Designed behavioral assays and conducted *in vivo* 2-photon imaging in *Drosophila*.
- Prototyped and fabricated a custom biomedical testing device using laser-cutting and CAD tools, addressing a gap in commercially available instrumentation.

**Visiting Student, Laboratory of Robert Langer, PhD •** Koch Institute, MIT; Cambridge, MA
Nov 2019 – May 2020

- Conducted *in vivo* efficacy studies for novel drug delivery systems in murine models, monitoring pharmacodynamic biomarkers and therapeutic outcomes.
- Validated therapeutic targets using high-dimensional flow cytometry and HPLC, ensuring rigorous assessment of drug mechanisms.

## Competitive Award

**NIH T32 Trainee**, Neurobiology of Information Storage Training Program (NISTP) • Awarded Sep. 1, 2022

- Competitively selected as a funded trainee under the National Institute of Mental Health (NIMH; grant #2T32MH067564).
- Authored the research proposal titled "Role of striatal µ-ORs in striatal coding of compulsive opioid use", securing funding based on scientific merit and experimental design.

## Select Publications and Abstracts

- **Wu, X.**; Fleps, S.W.; Kane, A.; Anair, J.D.; Centeno, M.V.; Apkarian, A.V.; Parker, J.G.
  The role of striatal µ-opioid receptors in opioid-driven striatal dynamics and behavior.
  *Manuscript in preparation.*
- Alsaiari, S.K.; Eshaghi, B.; Du, B.; Kanelli, M.; Li, G.; **Wu, X.**; Zhang, L.; Chaddah, M.; Lau, A.; Yang, X.; Langer, R. & Jaklenec, A.
  CRISPR–Cas9 delivery strategies for the modulation of immune and non-immune cells.
  *Nat Rev Mater* **10**, 44–61 (2024).
- Fleps, S.W.; Yang, B.; Moya, N.A.; **Wu, X.**; Yun, S. & Parker, J.G.
  Lateralized nigrostriatal dopamine pathway activation promotes early reversal learning.
  *Front. Behav. Neurosci*. 19:1703094 (2025).
- Potjer, E.; **Wu, X.**; Kane, A.; Parker, J.G.
  Parkinsonian striatal acetylcholine dynamics are refractory to L-DOPA treatment.
  *bioRxiv 2025.11.06.686401.*
- Conference: Gordon Research Conference Basal Ganglia (2024). Society for Neuroscience (2024).

## Community Involvement

**ETHOS Facilitator** • Loyola University Chicago, Chicago, IL
Jan 2026 – May 2026

- Facilitate after-school workshops in local high school to disseminate basic neuroscience knowledge and ethics to students.

**Graduate Student Mentor**, Summer Undergraduate Research Grants (SURG) Program • Northwestern University, Chicago, IL
July 2024 – present

- Mentored undergraduate in mouse surgical/behavioral experiments, data analysis, and scientific communication.